

Exercise Sheet 7 - Solutions

Predicate Logic – Syntax

1.
 - $\exists y. \exists z. \text{Student}(y, x) \wedge (\text{Module}(z, \text{Math}) \vee \text{Module}(z, \text{OOP})) \wedge \text{Enroll}(y, z)$
 - $\exists u. \exists v. \exists w. \text{Student}(u, x) \wedge \text{Student}(v, y) \wedge \text{Enroll}(u, w) \wedge \text{Enroll}(v, w)$
2. For example, let the domain be the set of propositional logic formulas. Let the signature be such that it includes at least the binary predicate symbol $\iff$, which stands for the logical equivalence relation on propositions (we will use infix notation). We can capture that $\iff$ is an equivalence relation as follows:
 - reflexive: $\forall x. x \iff x$
 - symmetric: $\forall x. \forall y. (x \iff y) \rightarrow (y \iff x)$
 - transitive: $\forall x. \forall y. \forall z. (x \iff y) \rightarrow (y \iff z) \rightarrow (x \iff z)$
3. Let the domain be D . Let the signature be
 - function symbols: $0, 1$ (arity 0), $-$ (arity 1), $+, \times$ (arity 2)
 - predicate symbols: $=$ (arity 2)

We will allow using infix notation for function and predicate symbols. The ring laws can be specified as follows:

- $\forall x. x + 0 = x$
 - $\forall x. \forall y. (x + y) = (y + x)$
 - $\forall x. x \times 1 = x$
 - $\forall x. \forall y. x \times y = y \times x$
 - $\forall x. x + (-x) = 0$
 - $\forall x. \forall y. \forall z. (x + y) + z = x + (y + z)$
 - $\forall x. \forall y. \forall z. (x \times y) \times z = x \times (y \times z)$
 - $\forall x. \forall y. \forall z. x \times (y + z) = (x \times y) + (x \times z)$
4.
 - $\forall x. \forall y. x > y \rightarrow x \geq (y + 1)$
 - $\forall x. \text{prime}(x) \rightarrow x > 1 \wedge \neg \exists y. \exists z. y > 1 \wedge z > 1 \wedge x = y \times z$

5. Here is a constructive Natural Deduction proof of

$$\begin{array}{c}
 (A \wedge \neg B) \rightarrow (A \rightarrow C) \rightarrow (D \rightarrow B) \rightarrow (\neg D \wedge C) \\
 \frac{\frac{\frac{\overline{D \rightarrow B}^3 \quad \overline{D}^4}{B} [\rightarrow E] \quad \frac{\overline{A \wedge \neg B}^1}{\neg B} [\wedge E_R]}{\frac{\perp}{\neg D}^4 [\neg I]} [\neg E] \quad \frac{\overline{A \rightarrow C}^2 \quad \frac{\overline{A \wedge \neg B}^1}{A} [\wedge E_L]}{C} [\rightarrow E]} \\
 \frac{\neg D \wedge C}{(D \rightarrow B) \rightarrow (\neg D \wedge C)}^3 [\wedge I] \\
 \frac{(D \rightarrow B) \rightarrow (\neg D \wedge C)}{(A \rightarrow C) \rightarrow (D \rightarrow B) \rightarrow (\neg D \wedge C)}^2 [\rightarrow I] \\
 \frac{(A \rightarrow C) \rightarrow (D \rightarrow B) \rightarrow (\neg D \wedge C)}{(A \wedge \neg B) \rightarrow (A \rightarrow C) \rightarrow (D \rightarrow B) \rightarrow (\neg D \wedge C)}^1 [\rightarrow I]
 \end{array}$$